RoboDev

STM32G431 + ESP32-C3 Integrated Robotics Prototyping Board

Design Specification | v0.8 | May 2026

Status	Schematic substantially complete — PCB layout not yet started
MCU	STM32G431RBTx (primary) + ESP32-C3-MINI-1-N4 (WiFi co-processor)
Repo	`https://github.com/alexbnz-1/RoboDev`

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Revision History

Version	Date	Summary of Changes
v0.1	Mar 2026	Initial document from notebook notes (22 Oct – 17 Nov 2025).
v0.2	Mar 2026	DRV8245-Q1 → DRV8874. MPU-6050 → BMI270.
v0.3	Mar 2026	Soldered WROOM module → ESP32-S3 DevKitC-1 v1.1 in dual-row headers.
v0.4	Mar 2026	NEMA 17 stepper support: 2× TMC2209, GPIO26–31 allocated.
v0.5	Mar 2026	MPU-6050 GY-521 replaces BMI270. ADS1115, PCF8574, OLED header. Servos 8→16. SPI/UART/carrier USB-C removed. 5× ADC1 headers.
v0.6	Mar 2026	CAN bus (TWAI, GPIO7/32, SN65HVD230). Encoders 4→2. nFAULT→indicator LED. UART1 header (GPIO47/21). GPIO38/48 written off. Separate Vmot-DC and Vstep rails. 0 spare GPIOs. BOM ~NZ$82.81.
v0.7	Mar 2026	⋆ **I^{2}C expansion array:** 2× TCA9548A mux (0x70/0x71) + 6× Qwiic JST SH + 10× 2.54 mm 4-pin headers = 16 independent expansion ports. Reverse-ARP HAL for automatic peripheral discovery. BOM ~NZ$90–92.
v0.8	May 2026	⋆ **Dual-MCU architecture:** STM32G431RBTx replaces ESP32-S3 as primary MCU. ESP32-C3-MINI-1-N4 added as SPI-attached WiFi/BT co-processor. BMI270 IMU moves to SPI (co-processor sheet). USB4085-GF-A Type-C on carrier. SWD debug port added. XT60PW-M replaces XT30 power connector. IRF4905 PMOS reverse protection. 74HC595 shift register added. XL4016 ×2 for VMOT + VSTEP. KiCad schematic implementation begun (all major blocks drawn).

1 Introduction

RoboDev is an open-source integrated robotics prototyping board built around a **dual-MCU architecture**: an STMicroelectronics STM32G431RBTx Cortex-M4F as the primary motor control processor, and an Espressif ESP32-C3-MINI-1-N4 module as a SPI-attached WiFi and Bluetooth co-processor. It is designed to eliminate the wiring overhead that typically accompanies microcontroller-based robotics projects by integrating motor drivers, a servo PWM controller, stepper motor drivers, a CAN bus transceiver, an IMU, analogue ADC expanders, digital GPIO expansion, an I^{2}C expansion array, and a complete multi-rail power supply onto a single carrier PCB.

The core philosophy of RoboDev is that only three types of wire should ever need to be connected to the board: power in, and wires out to sensors and actuators. All signal routing — from the MCU to driver ICs, from the I^{2}C bus to peripherals, from the power switch to every rail — is handled on the PCB itself.

1.1 Motivation

Robotics prototyping typically requires a breadboard or custom wiring harness to connect motor driver breakout boards, servo controllers, and sensor modules. This introduces noise susceptibility, mechanical fragility, and significant bring-up time before any firmware can be tested. RoboDev solves this by providing a production-ready carrier board with the STM32G431RBTx at the centre — a Cortex-M4F processor from ST's G4 series that is purpose-built for motor control and digital power conversion, with hardware PWM timers, FDCAN, and a hardware FPU.

Wireless connectivity is provided by the ESP32-C3-MINI-1-N4, mounted on the carrier and connected to the STM32 via SPI. This separation of concerns keeps the hard-real-time motor control tasks on the deterministic STM32 while offloading WiFi, Bluetooth, and any cloud connectivity tasks to the ESP32-C3.

1.2 Design Philosophy

- Full-featured and self-contained — no daughterboards or breakout modules required
- Dual-MCU architecture: STM32G431RBTx for real-time control; ESP32-C3-MINI-1-N4 for wireless
- Any motor type supported simultaneously: 4× brushed DC, 16× servo, 2× NEMA 17 stepper, and CAN-capable motor controllers
- BMI270 6-axis IMU on SPI for high-speed attitude estimation
- Three independently adjustable power rails for motors, steppers, and servos
- 16-port I^2C expansion array via 2× TCA9548A mux (zero GPIO cost)
- All components hand-solderable — no BGA or ultra-fine-pitch packages
- Open source: schematics, PCB layout, firmware, and documentation on GitHub

1.3 Target Applications

- 2WD and 4WD wheeled robot platforms
- Multi-axis robot arms with CAN-capable joint controllers
- Mechatronics rigs requiring mixed motor types on a single controller
- Autonomous vehicles with wireless telemetry and IMU sensor fusion
- Sensor-dense robots using the I^2C expansion array for multiple sensors
- Educational robotics platforms requiring a robust, well-documented base board

2 System Overview

RoboDev uses a two-MCU design. The **STM32G431RBTx** is the primary processor, handling all real-time motor control, CAN communication, sensor interfacing, and I^2C bus management. The **ESP32-C3-MINI-1-N4** sits alongside it on the carrier PCB, connected to the STM32 via SPI, and handles all WiFi and Bluetooth tasks.

The board is divided into eight functional domains: power management, primary MCU (STM32), WiFi co-processor (ESP32-C3), motor control (DC, stepper, servo), communication (I^2C, CAN, UART), sensing (IMU, analogue, digital), expansion (Qwiic, GPIO headers), and the I^2C expansion array.

2.1 Core Requirements Summary

Category	Requirement / Specification
Primary MCU	STM32G431RBTx · Cortex-M4F · 170 MHz · FPU · 128 KB Flash · LQFP-64
WiFi Co-processor	ESP32-C3-MINI-1-N4 · SPI-connected · 802.11b/g/n + Bluetooth 5
DC Motors	4 channels · DRV8874PWPR · 4.5–37 V · 2.1 A cont. / 6 A peak · nFAULT LED

Category	Requirement / Specification
Servos	16 channels · PCA9685PW I^2C PWM driver (0x40) · 12-bit resolution · 5 V servo rail
Stepper Motors	2 channels · TMC2209-LA-T · NEMA 17 · 2 A RMS / 2.8 A peak · UART-configurable · StealthChop2
CAN Bus	FDCAN (ISO 11898-1) · SN65HVD230D 3.3 V transceiver · up to 1 Mbps · 5.08 mm screw terminal
IMU	BMI270 · 6-axis accel + gyro · SPI interface · on co-processor sheet
Quadrature Encoders	2 channels · 4-pin 2.54 mm headers
Native Analogue ADC	3× 2.54 mm headers · STM32 ADC inputs
I^2C Analogue ADC	4 channels · ADS1115IDGST (0x48) · 16-bit · I^2C
GPIO Expander	PCF8574DW (0x20) · 8-channel digital I/O · I^2C
I^2C Expansion Array	16 independent ports · 2× TCA9548APWR mux (0x70/0x71) · 3× Qwiic + 13× 2.54 mm · zero GPIO cost
USB	Type-C · USB4085-GF-A · on STM32 sheet
SWD Debug	2.54 mm header · STM32 programming and debug
UART Headers	2× 2.54 mm (debug + general purpose)
Shift Register	74HC595 · 8-bit serial-in parallel-out · SPI-driven (ESP32-C3 sheet)
Power Input	XT60PW-M · reverse polarity protection (IRF4905 PMOS) · slide switch

3 Power Architecture

The power system is designed around four independent rails, each serving a distinct subsystem. Separating VMOT from VSTEP allows brushed DC motors and stepper motors to be run at different voltages simultaneously. All rails are derived from a single protected XT60 input, switched by a common high-current slide switch.

3.1 Power Input and Protection

Power enters the board via an XT60PW-M connector. Input voltage should be at least 6 V to allow headroom for the buck converter generating the 5 V logic rail. Do not exceed 36 V (the rated ceiling of the XL4016 buck converters).

An IRF4905 P-channel power MOSFET provides reverse voltage protection on VIN before the power switch. MBR2045CT Schottky diodes are used for power path management. An SS14 diode handles USB/power path arbitration.

Warning: A high-current slide switch (OS102011MS2QN1) is placed after the reverse voltage protection circuit. It switches all rails simultaneously. When laying out the PCB, ensure the switch trace width is adequate for the total system current.

3.2 Power Rails

Rail	Output	Current	Load	Key Component / Notes
VMOT	Adj. 4.5–24 V	5 A+ cont.	DC Motors M1–M4	XL4016 step-down · trimmer pot on FB · 470 µF bulk ×4
VSTEP	Adj. 4.5–24 V	3 A+ cont.	Steppers S1–S2	XL4016 step-down · independent rail
+5 V	Fixed 5 V	3 A+ cont.	Servo rail, USB, ESP32-C3	LM2596T-5 synchronous buck · 33 µH inductor
+3.3 V	Fixed 3.3 V	1 A+ cont.	STM32, all ICs, sensors	AP2112K-3.3 LDO from 5 V · all I^2C devices

Warning: VMOT and VSTEP are independent adjustable XL4016 buck converters. Each has its own trimmer potentiometer. Label both clearly on the PCB silkscreen to prevent accidental cross-connection.

3.3 Bulk Decoupling Capacitors

Each DRV8874 DC motor driver requires at least one electrolytic bulk capacitor rated for the VMOT voltage placed as close as possible to the VM pin. A minimum of 470 µF at 50 V is specified per driver. Each TMC2209 stepper driver similarly requires bulk capacitance on the VSTEP rail. All ICs additionally receive 100 nF ceramic decoupling capacitors.

4 Microcontrollers

4.1 Primary MCU — STM32G431RBTx

The STM32G431RBTx (U_MCU1) is a 64-pin LQFP Cortex-M4F microcontroller from STMicroelectronics' G4 series. Running at 170 MHz with a hardware FPU, it is purpose-built for motor control and digital power conversion applications.

Note: The schematic sheet is labelled `stm32` and uses file `esp32.kicad_sch` — a legacy filename from the earlier ESP32-S3 design. The actual placed component is STM32G431RBTx.

Key peripherals used on this board:

- **Timers (TIM/HRTIM)** — motor PWM generation for M1–M4 DC motor channels and S1–S2 stepper STEP pulses
- **FDCAN** — hardware CAN controller via SN65HVD230D transceiver
- **I^2C** — shared bus for PCA9685, ADS1115, PCF8574, TCA9548A ×2
- **SPI** — connection to ESP32-C3-MINI-1-N4 co-processor and 74HC595 shift register
- **UART/USART** — TMC2209 UART configuration interface + debug/general UART headers
- **ADC** — native ADC inputs broken out to 3× 2.54 mm headers
- **USB FS** — via USB4085-GF-A Type-C connector
- **SWD** — programming and debug via 2.54 mm header (J_SWD1)

The BOOT0 pin is pulled to GND via a 10 kΩ resistor with a MJTP1230 push-button to allow entry into the DFU bootloader. A separate MJTP1230 provides system RESET. A 100 nF capacitor is placed on the $\overline{\text{NRST}}$ pin.

4.2 WiFi Co-processor — ESP32-C3-MINI-1-N4

The ESP32-C3-MINI-1-N4 (on the `spi_bus` sheet) provides 802.11b/g/n WiFi and Bluetooth 5 LE connectivity. It is connected to the STM32 via SPI, with the following interface signals:

- **SPI_SCK, SPI_MOSI, SPI_MISO** — SPI bus to/from STM32
- **SPI_CS** — chip select from STM32
- **SPI_LATCH** — latch signal (also drives 74HC595 shift register)
- **WIFI_IRQ** — interrupt output from ESP32-C3 to STM32

The ESP32-C3 also connects to the BMI270 IMU and the 74HC595 shift register on its sheet.

4.3 74HC595 Shift Register

A 74HC595 8-bit serial-in parallel-out shift register is placed on the `spi_bus` sheet alongside the ESP32-C3. It is clocked via the SPI bus (or the SPI_LATCH signal), providing 8 additional digital output lines without consuming extra GPIO on the STM32. Typical use cases include status LEDs, output expansion, and simple digital control signals.

5 Signal Interface Summary

The following table lists the hierarchical sheet pin names that cross between sheets in the KiCad schematic. Exact STM32 pin numbers are TBD (to be verified during PCB layout).

Signal	Dir.	Sheet	Function
M1_PWM / M1_DIR	Out	stm32 → motor_dc	DC motor M1 PWM + direction
M2_PWM / M2_DIR	Out	stm32 → motor_dc	DC motor M2 PWM + direction
M3_PWM / M3_DIR	Out	stm32 → motor_dc	DC motor M3 PWM + direction
M4_PWM / M4_DIR	Out	stm32 → motor_dc	DC motor M4 PWM + direction
S1_STEP/DIR/EN	Out	stm32 → motor_stepper	Stepper S1 control
S2_STEP/DIR/EN	Out	stm32 → motor_stepper	Stepper S2 control
TMC_UART_TX/R	Bidir	stm32 ↔ motor_stepper	TMC2209 UART config bus
I2C_SCL/SDA	Out	stm32 → i2c_bus, motor_servo	Shared I^{2}C bus
SPI_CS/MOSI/MISO	Bidir	stm32 ↔ spi_bus	SPI to ESP32-C3 + 74HC595
SPI_LATCH	Out	stm32 → spi_bus	74HC595 latch
WIFI_IRQ	In	spi_bus → stm32	ESP32-C3 interrupt
IMU_INT1	In	spi_bus → stm32	BMI270 interrupt
GPIO_INT	In	i2c_bus → stm32	PCF8574 interrupt
IPROPI_M1–M4	In	motor_dc → i2c_bus	DRV8874 current sense to ADS1115
VMOT	Pwr	power → motor_dc	DC motor power rail

Signal	Dir.	Sheet	Function
VSTEP	Pwr	power → motor_stepper	Stepper power rail
VSERVO	Pwr	power → motor_servo	Servo power rail
VBUS / ADC_VIN	Pwr	power → stm32	USB bus power + ADC reference

6 Motor Control

6.1 DC Brushed Motors — DRV8874PWPR × 4

Four DRV8874PWPR H-bridge motor driver ICs (U_M1–U_M4) provide control for four independent brushed DC motor channels (M1–M4). The DRV8874 is a fully integrated H-bridge with hardware overcurrent protection, thermal shutdown, and undervoltage lockout.

Parameter	Value / Detail
Part number	DRV8874PWPR (HTSSOP-16)
Supply voltage (VM)	4.5 V – 37 V (VMOT rail)
Continuous current	2.1 A per channel
Peak current	6 A per channel
Control M1	M1_PWM (PWM/Phase), M1_DIR (DIR/Enable) from STM32
Control M2	M2_PWM, M2_DIR
Control M3	M3_PWM, M3_DIR
Control M4	M4_PWM, M4_DIR
Output connector	2-pin 5.08 mm screw terminal per channel
nFAULT handling	All 4 nFAULT pins wired-OR open-drain → red indicator LED (LTST-C190KRKT). No MCU GPIO consumed.
Bulk capacitance	470 µF 50 V electrolytic per driver (C_VM1–C_VM4), close to VM pin
Current sense	IPROPI output routed to ADS1115 (A6–A9) for per-motor current monitoring

6.2 NEMA 17 Stepper Motors — TMC2209-LA-T × 2

Two TMC2209-LA-T silent stepper motor driver ICs (U_S1–U_S2) drive two NEMA 17 bipolar stepper motors from the independent VSTEP rail. The TMC2209 features StealthChop2 for near-silent operation and SpreadCycle for high-torque applications, configurable via UART.

Parameter	Value / Detail
Part number	TMC2209-LA-T (QFN-28)
Supply voltage (VM)	4.5 V – 36 V (VSTEP rail — independent from VMOT)
RMS current	2.0 A RMS per channel
Peak current	2.8 A peak per channel
Control S1	S1_STEP, S1_DIR, S1_EN (active LOW)
Control S2	S2_STEP, S2_DIR, S2_EN (active LOW)
EN pull-up	10 kΩ to 3.3 V — drivers disabled by default at power-on

Parameter	Value / Detail
UART config	TMC_UART_TX/RX to STM32 (optional per-application current tuning)
Bulk capacitance	100 µF 35 V+ per driver

Note: The VSTEP rail is intentionally separate from VMOT. This allows, for example, 12 V NEMA 17 operation alongside 6 V DC gearmotors simultaneously.

6.3 Servo Motors — PCA9685PW × 16

A PCA9685PW 16-channel 12-bit PWM driver IC provides servo control for up to 16 hobby servo channels simultaneously via I^2C. All 16 PWM outputs are broken out to 3-pin 2.54 mm servo headers carrying GND, VSERVO, and PWM signal.

Parameter	Value / Detail
Part number	PCA9685PW (TSSOP-28)
I^2C address	0x40 (ADDR pins → GND)
PWM resolution	12-bit (4096 steps)
PWM frequency	Configurable 24 Hz – 1526 Hz (typically 50 Hz for standard servos)
Servo voltage	VSERVO (derived from +5 V rail via MBR2045CT protection diode)
Servo headers	3-pin 2.54 mm GND / VSERVO / PWM (16 channels)
Control interface	I^2C via STM32 I^2C bus

6.4 Motor Compatibility

Motor Type	Compatible	Notes
TT yellow gearmotors (2WD/4WD)	✓	Common chassis motors — ideal fit
Pololu micro/mini DC gearmotors	✓	Well within DRV8874 2.1 A continuous rating
25 mm / 37 mm DC gearmotors	✓*	Stall may approach 6 A peak — current limiting recommended
775-size DC motors	×	Exceeds DRV8874 continuous rating — use external driver
Hobby servos (3–8.4 V)	✓	PCA9685 servo rail, up to 16 channels simultaneously
NEMA 17 stepper motors	✓	TMC2209 StealthChop2 · up to 2 A RMS / 2.8 A peak
CAN motor controllers (VESC, ODrive)	✓	FDCAN via SN65HVD230D · ISO 11898-1

7 Communication Interfaces

7.1 I²C Bus

A single I²C bus from the STM32 carries all on-board I²C peripherals plus the TCA9548A expansion array. Pull-up resistors of 4.7 kΩ to 3.3 V are provided on-board.

Device	Addr.	Function	Notes
PCA9685PW	0x40	16-ch servo PWM driver	Main bus — always visible
ADS1115IDGST	0x48	4-ch 16-bit ADC	Main bus — ADDR→GND
PCF8574DW	0x20	8-ch digital GPIO expander	Main bus — INT to STM32 (GPIO_INT)
OLED display	0x3C	0.96 " display (user-supplied)	4-pin 2.54 mm header (J_OLED1)
TCA9548APWR Mux #1	0x70	I²C mux — ports 0–7	Main bus — always visible
TCA9548APWR Mux #2	0x71	I²C mux — ports 8–15	Main bus — always visible
Expansion 0–15	Any	User I²C devices	Behind mux — no address conflicts

Note: The BMI270 IMU is on **SPI**, not I²C. It is located on the `spi_bus` sheet alongside the ESP32-C3. The main I²C bus has seven on-board devices; no address conflicts exist between them.

7.2 CAN Bus (FDCAN)

CAN bus support is implemented using the STM32G431's built-in FDCAN controller, conforming to ISO 11898-1. An SN65HVD230D 3.3 V CAN transceiver provides the physical layer.

Parameter	Value / Detail
Protocol	ISO 11898-1 (CAN 2.0A/B / CAN FD capable) via STM32 FDCAN hardware
Transceiver	SN65HVD230D (SOIC-8, 3.3 V compatible, up to 1 Mbps)
Bus connector	5.08 mm 2-pin screw terminal (J_CAN1) — CANH / CANL
Termination	120 Ω resistor (R_CAN_TERM1) on board
Logic level	3.3 V — directly compatible with STM32G431 GPIO
Use cases	CAN motor controllers (VESC, ODrive), robot arm joints, inter-board comms

Warning: Do not substitute the SN65HVD230D with a 5 V transceiver such as the MCP2551. The STM32G431 GPIO pins are 3.3 V tolerant; a 5 V transceiver output will permanently damage the MCU.

7.3 USB (Type-C)

A USB4085-GF-A USB Type-C connector (X1) on the STM32 sheet provides USB Full-Speed connectivity directly to the STM32G431's USB FS peripheral. This is used for firmware programming (DFU bootloader via BOOT0 button) and virtual COM port (VCP) communication.

7.4 SWD Debug Interface

A 4-pin 2.54 mm SWD header (J_SWD1) exposes the STM32G431's Serial Wire Debug interface (SWDIO / SWDCLK / GND / 3.3 V). This supports programming and live debugging via ST-LINK, J-Link, or compatible probe.

7.5 UART Headers

- **J_UART_DBG1** — UART debug port (4-pin 2.54 mm). Intended for serial monitor and low-level firmware diagnostics.
- **J_UART2** — General-purpose UART header (4-pin 2.54 mm). Use for GPS (NMEA), LiDAR (RPLIDAR, TF-Luna), or any UART-based peripheral.

8 Sensing and Analogue Inputs

8.1 Inertial Measurement Unit — BMI270

The BMI270 6-axis IMU is placed on the `spi_bus` sheet alongside the ESP32-C3-MINI-1-N4. It provides 3-axis accelerometer and 3-axis gyroscope data via SPI. An interrupt output (IMU_INT1) is routed to the STM32 for data-ready notification.

- Interface: SPI
- Interrupt: IMU_INT1 → STM32
- Firmware: BMI270 SPI driver recommended; Madgwick or Mahony filter for attitude estimation

8.2 Native Analogue Inputs — ADC Headers

Three 2.54 mm headers (J_ADC1, J_ADC2, J_ADC3) break out STM32G431 ADC input channels. Exact GPIO assignments are to be determined during schematic finalisation and PCB layout.

- Pinout: GND / 3.3 V / SIG (3-pin headers)
- Resolution: 12-bit (STM32 ADC1/ADC2)

8.3 I^{2}C Analogue Inputs — ADS1115IDGST

The ADS1115IDGST 16-bit I^{2}C ADC (U_ADC1) provides four additional analogue channels (A6–A9) at significantly higher resolution than the native ADC. The DRV8874 IPROPI current sense outputs are routed here for per-motor current monitoring.

- I^{2}C address: 0x48 (ADDR→GND)
- Resolution: 16-bit (±32767 counts), programmable gain: ±0.256 V to ±6.144 V FSR
- IPROPI inputs: IPROPI_M1–M4 from DRV8874 → ADS1115 (via i2c_bus sheet)

8.4 PCF8574DW — Digital GPIO Expander

The PCF8574DW (U_GPIO1) provides 8 additional digital I/O lines via I^{2}C at address 0x20. An INT signal is routed to the STM32 (GPIO_INT) for interrupt-driven pin change notification. A pull-up resistor (R_GPIO_INT1) is placed on the INT line.

8.5 Quadrature Encoders

Two quadrature encoder channels are supported. Each channel provides a 4-pin 2.54 mm header (J_ENC1, J_ENC2) with pinout GND / 3.3 V / Chan-A / Chan-B.

8.6 GPIO Header

A general-purpose GPIO header (J_GPIO1) breaks out additional STM32 GPIO pins for user-defined digital I/O.

9 I²C Expansion Array

Two TCA9548APWR 8-channel I²C multiplexers (U_MUX1 at 0x70, U_MUX2 at 0x71) are added to the main I²C bus. Each provides 8 independently switched downstream channels, giving 16 expansion ports total. Each port is a fully isolated I²C segment — devices on different ports can share the same I²C address without conflict.

9.1 Expansion Port Layout

Port	Connector	Mux Channel	Isolation	Notes
0	QWIIC JST SH 4-pin	TCA9548A #1 Ch0	Independent	QWIIC port 1 on Mux #1
1	QWIIC JST SH 4-pin	TCA9548A #1 Ch1	Independent	QWIIC port 2 on Mux #1
2	QWIIC JST SH 4-pin	TCA9548A #1 Ch2	Independent	QWIIC port 3 on Mux #1
3	2.54 mm 4-pin	TCA9548A #1 Ch3	Independent	GND/3.3 V/SDA/SCL
4	2.54 mm 4-pin	TCA9548A #1 Ch4	Independent	
5	2.54 mm 4-pin	TCA9548A #1 Ch5	Independent	
6	2.54 mm 4-pin	TCA9548A #1 Ch6	Independent	
7	2.54 mm 4-pin	TCA9548A #1 Ch7	Independent	
8	QWIIC JST SH 4-pin	TCA9548A #2 Ch0	Independent	QWIIC port 1 on Mux #2
9	2.54 mm 4-pin	TCA9548A #2 Ch1	Independent	
10	2.54 mm 4-pin	TCA9548A #2 Ch2	Independent	
11	2.54 mm 4-pin	TCA9548A #2 Ch3	Independent	
12	2.54 mm 4-pin	TCA9548A #2 Ch4	Independent	
13	2.54 mm 4-pin	TCA9548A #2 Ch5	Independent	
14	2.54 mm 4-pin	TCA9548A #2 Ch6	Independent	
15	2.54 mm 4-pin	TCA9548A #2 Ch7	Independent	

9.2 TCA9548A Architecture

Parameter	Value / Detail
Part number	TCA9548APWR (TSSOP-24)
I²C address Mux #1	0x70 (A2/A1/A0 = 0/0/0 → GND)
I²C address Mux #2	0x71 (A2/A1/A0 = 0/0/1 → Mux #2 A0 → 3.3 V)
Channels per IC	8 independent I²C downstream channels
Total expansion ports	16 (8 from Mux #1 + 8 from Mux #2)
Supply voltage	3.3 V
Pull-up resistors	4.7 kΩ on SDA/SCL of main bus only (mux outputs are active)

Parameter	Value / Detail
Address conflict risk	None between ports — each port is on an independently switched segment
Reset pull-ups	10 kΩ pull-ups on /RESET pins of both TCA9548A ICs

9.3 Reverse-ARP HAL — Automatic Peripheral Discovery

The firmware HAL implements a reverse-ARP discovery mechanism for the expansion array. On startup, the HAL iterates through all 16 expansion ports, temporarily enables each channel, performs an I^2C scan, records any device addresses found, and then disables the channel. The result is a port map that application code can query.

- On startup: scan all 16 ports → build port map → log device addresses per port
- At runtime: request `i2c_port_N` → HAL enables TCA9548A channel → performs transaction → disables channel
- No address conflicts possible — HAL enforces single-channel activation

Warning: Only one TCA9548A channel should be active at any time. The HAL enforces this. Do not manually write to the TCA9548A channel registers from application code while the HAL is active.

10 Connector Reference

Ref	Type	Qty	Pitch	Function / Pinout
J_XT60	XT60PW-M	1	XT60	VIN power input
J_CAN1	Screw terminal	1	5.08 mm 2-pin	CAN: CANH / CANL
J_ENC1, J_ENC2	Pin header	2	2.54 mm 4-pin	Encoders: GND / 3.3 V / Chan-A / Chan-B
J_UART_DBG1	Pin header	1	2.54 mm 4-pin	UART debug: GND / 3.3 V / TX / RX
J_UART2	Pin header	1	2.54 mm 4-pin	UART general: GND / 3.3 V / TX / RX
J_OLED1	Pin header	1	2.54 mm 4-pin	OLED: GND / 3.3 V / SDA / SCL
J_SWD1	Pin header	1	2.54 mm 4-pin	SWD: GND / 3.3 V / SWDIO / SWDCLK
J_ADC1–J_ADC3	Pin header	3	2.54 mm 3-pin	ADC: GND / 3.3 V / SIG
J_GPIO1	Pin header	1	2.54 mm 4-pin	Direct GPIO header
Servo headers	Pin header	16	2.54 mm 3-pin	Servo: GND / VSERVO / PWM
Motor terminals	Screw terminal	4	5.08 mm 2-pin	DC motor outputs M1–M4
Stepper headers	Pin header	2	2.54 mm 4-pin	STEP/DIR/EN per stepper
J_MUX1_CH0–CH7	Mixed	8	QWIIC / 2.54 mm	I^2C expansion ports 0–7
J_MUX2_CH0–CH7	Mixed	8	QWIIC / 2.54 mm	I^2C expansion ports 8–15

Ref	Type	Qty	Pitch	Function / Pinout
X1	USB Type-C	1	USB4085-GF-A	USB FS / DFU programming
H1–H4	Mounting hole	4	M3 pad	Grounded M3 mounting holes

11 Bill of Materials (v0.8)

Estimated mid-range unit pricing at Qty = 1. Excludes PCB assembly labour, shipping, and import duties. Rows highlighted in blue are new or significantly changed in v0.8.

#	Component	Part Number	Qty	Notes
1	STM32G431RBTx MCU	STM32G431RBTx (LQFP-64)	1	Primary MCU, 170 MHz Cortex-M4F
2	ESP32-C3-MINI-1-N4	ESP32-C3-MINI-1-N4	1	WiFi/BT co-processor, SPI interface
3	DRV8874 H-Bridge	DRV8874PWPR (HTSSOP-16)	4	DC motors M1–M4
4	SN65HVD230 CAN	SN65HVD230D (SOIC-8)	1	CAN bus transceiver
5	CAN screw terminal	5.08 mm 2-pin	1	CANH / CANL
6	nFAULT LED	LTST-C190KRKT Red	1	DRV8874 fault indicator
7	PCA9685 servo driver	PCA9685PW (TSSOP-28)	1	16-ch PWM servo
8	ADS1115 ADC	ADS1115IDGST (MSOP-10)	1	16-bit I²C ADC, motor current sense
9	PCF8574 GPIO expander	PCF8574DW (SOIC-16)	1	8-ch digital I/O expander
10	TCA9548A I²C Mux	TCA9548APWR (TSSOP-24)	2	Mux #1 (0x70), Mux #2 (0x71)
11	QWIIC header	JST SH SM04B-SRSS-TB	3	I²C expansion QWIIC ports
12	Expansion 2.54 mm headers	4-pin 2.54 mm	13	I²C expansion headers
13	BMI270 IMU	BMI270	1	6-axis IMU, SPI interface
14	74HC595 shift register	74HC595	1	8-bit serial output expander
15	USB Type-C connector	USB4085-GF-A	1	USB FS / DFU
16	XT60 power connector	XT60PW-M	1	Main power input
17	IRF4905 PMOS	IRF4905	1	Reverse polarity protection
18	MBR2045CT Schottky	MBR2045CT	1+	Power path protection
19	SS14 Schottky diode	SS14	1	USB power path
20	OS102011MS2QN1 switch	OS102011MS2QN1	1	Main power slide switch
21	MJTP1230 pushbutton	MJTP1230	2	BOOT0 + RESET buttons
22	XL4016 buck converter	XL4016	2	VMOT + VSTEP adjustable rails
23	LM2596T-5 buck	LM2596T-5	1	Fixed 5 V logic rail
24	AP2112K-3.3 LDO	AP2112K-3.3 (SOT-25)	1	Fixed 3.3 V logic
25	33 µH / 4 A inductor	–	1	5 V buck output filter

#	Component	Part Number	Qty	Notes
26	47 µH / 12 A inductor	–	2	VMOT / VSTEP buck output filters
27	Electrolytic cap 470 µF 50 V	–	3+	Motor/bulk capacitors
28	Electrolytic cap 680 µF 50 V	–	1	VIN bulk input cap
29	Electrolytic cap 220 µF 25 V	–	1	5 V rail bulk cap
30	Bourns 3296W trimmer 10 kΩ	Bourns 3296W	2+	VMOT / VSTEP feedback dividers
31	100 nF ceramic 0603 (×25)	GRM188R71C104	25	IC bypass decoupling
32	Resistor assortment 0603 (×40)	Misc.	40	Pull-ups, dividers, LED limiting
33	2-layer PCB ~130×95 mm	JLCPCB / PCBWay	1	Design TBD

12 Recommended Improvements

Improvement	Priority	Notes
Battery voltage monitor	High	Resistor divider on VIN → any ADC header. Zero GPIO cost.
Per-motor IPROPI current sense	High	DRV8874 IPROPI → ADS1115. Already routed in schematic (IPROPI_M1–M4).
LiPo XT60 connector	Medium	XT60PW-M already placed. Consider adding balancer connector header.
On-board buzzer	Low	Requires an available GPIO. Assign from freed signals during layout.
WS2812B RGB status LED	Low	ESP32-C3 has spare GPIOs — can drive RGB LED without loading STM32.
CAN bus termination option	Medium	120 Ω R_CAN_TERM1 already placed. Add solder-bridge jumper to enable/disable.

13 Open Questions and Next Steps

13.1 Verification Required

- Confirm all STM32G431RBTx pin assignments in schematic (ESP32-S3 pin table is obsolete)
- Prototype I^2C bus — verify all 6 on-board device addresses and pull-up values at full loading
- Prototype CAN bus — verify SN65HVD230D wiring and FDCAN firmware configuration (bitrate, mode)
- Verify TCA9548A address selection resistors for 0x70 (all pins→GND) and 0x71 (A0→3.3 V)
- Test SPI interface between STM32 and ESP32-C3 — confirm protocol, clock speed, and IRQ behaviour
- Confirm BMI270 SPI routing and interrupt (IMU_INT1) behaviour

13.2 KiCad / PCB Design Tasks

- Review and clean up `esp32.kicad_sch` — rename sheet from "stm32" to something less confusing; rename file from `esp32.kicad_sch` if desired
- Populate the empty `connectors.kicad_sch` sheet or remove it from the hierarchy
- Assign and verify all footprints in the KiCad project
- Generate netlist and begin PCB layout
- Separate ground planes: high-current motor ground physically isolated from logic ground with single star-point
- Ensure VMOT and VSTEP rails are clearly labelled on silkscreen
- Plan PCB size to accommodate all expansion port headers (∼130×95 mm estimated)
- Add solder-bridge jumper for CAN 120 Ω termination
- Mark BOOT0 / RESET buttons clearly on silkscreen

13.3 Firmware

- STM32 HAL/LL bringup: I^2C scan (all 16 expansion ports), PWM test, CAN loopback, UART echo
- TMC2209 UART configuration driver for per-application current tuning
- BMI270 SPI driver + Madgwick / Mahony filter for attitude estimation
- ESP32-C3 SPI bridge firmware — define message protocol to/from STM32
- Implement reverse-ARP HAL for TCA9548A expansion array
- Define HAL API: `i2c_port_scan_all()`, `i2c_port_read(port, addr, reg, len)`, `i2c_port_write(po`
`addr, reg, data, len)`
- Generate full LCSC / Mouser priced BOM from this document for procurement

A I^2C Address Map

Address	Device	Bus Location	Notes
0x20	PCF8574DW	Main bus (fixed)	8-ch digital GPIO expander
0x3C	OLED display	Main bus (fixed)	0.96 ” OLED header — user-supplied
0x40	PCA9685PW	Main bus (fixed)	16-ch servo PWM driver
0x48	ADS1115IDGST	Main bus (fixed)	4× 16-bit analogue ADC
0x70	TCA9548APWR Mux #1	Main bus (fixed)	8-ch I^2C mux — ports 0–7
0x71	TCA9548APWR Mux #2	Main bus (fixed)	8-ch I^2C mux — ports 8–15
Various	User devices	Expansion ports 0–15	No conflicts between ports

Not on I^2C: BMI270 (SPI, spi_bus sheet) — *no longer on the I^2C bus.*

B Schematic Sheet Structure

Sheet Name	File	Contents
Root	`Robodev.kicad_sch`	Top-level interconnect, mounting holes (H1–H4)

Sheet Name	File	Contents
power	`power.kicad_sch`	XT60 input, IRF4905 reverse protect, slide switch, XL4016×2, LM2596T-5, AP2112K-3.3, all bulk caps
stm32	`esp32.kicad_sch`	STM32G431RBTx, SN65HVD230D CAN, USB4085-GF-A, SWD header, UART headers, ADC headers, encoder headers, GPIO header, BOOT0/RESET buttons
spi_bus	`spi_bus.kicad_sch`	ESP32-C3-MINI-1-N4, BMI270 IMU, 74HC595 shift register
i2c_bus	`i2c_bus.kicad_sch`	TCA9548APWR×2 muxes, ADS1115IDGST ADC, PCF8574DW GPIO expander, all I^{2}C expansion headers (QWIIC + 2.54 mm), OLED header
motor_dc	`motor_dc.kicad_sch`	DRV8874PWPR×4, motor screw terminals, nFAULT LED, bulk capacitors
motor_stepper	`motor_stepper.kicad_sch`	TMC2209-LA-T×2, stepper output headers
motor_servo	`motor_servo.kicad_sch`	PCA9685PW, 16× servo headers
connectors	`connectors.kicad_sch`	(Currently empty — reserved for future use)